Biology 2
0510

GENERAL CERTIFICATE OF EDUCATION BOARD

General Certificate of Education Examination

JUNE 2026 **ORDINARY LEVEL**

Subject Title	**BIOLOGY**
Paper No.	**2**
Subject Code No.	**0510**

Duration: Two and Half Hours

*Answer **FIVE** questions, **ALL THREE (3)** from **SECTION A** and any **TWO (2)** from **SECTION B***

All questions carry 20 marks each. For your guidance, the approximate mark for each part of a question is indicated in brackets.

Begin each question on a new page.

Give labelled diagrams where these will make the answers clearer.

You are advised to read carefully through the question paper before you begin your answers.

You are reminded of the necessity for good English and orderly presentation in your answers.

In calculations, you are advised to show all the steps in your working, giving your answer at each stage.

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SECTION A
Answer ALL 3 questions

1. **(a)** (i) List eight (8) characteristics common to insects. **(4,5 marks)**
(ii) Draw a large labelled diagram of a dipteran fly. **(5 marks)**

(b) Give five (5) reasons why insects are highly successful in their habitat. **(6 marks)**

(c) Insects undergo either complete or incomplete metamorphosis in their life cycles. Differentiate between these two developmental cycle. **(Total = 20 marks)**

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2. **(a)** List the environmental factors that affect the rate of transpiration. **(5 marks)**

(b) How is water in the cortex of roots of plants made available for photosynthesis in the leaves? **(7 marks)**

(c) How would you measure the rate of uptake of water by a leafy shoot in a bright sunny day? **(8 marks)**
(Total = 20 marks)

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3. **(a)** Define the following terms as used in Biotechnology.
(i) Plasmid.
(ii) Vector.
(iii) Ligase.
(iv) Bacteriophage
(v) Clonc. **(2,2,1,1,1,2 marks)**

(b) Differentiate between a food chain and a food web. **(6 marks)**

(c) In an abandoned school farm, the following organisms were discovered to be living there after a period of three years: hawks, green vegetation such as huckle berry and cabbage, nymphs of grasshopper, flying grasshoppers, toads under green vegetation, earthworms under rotten logs of wood and lizards. Using the above organisms, construct a suitable food web to establish the relationship between these organisms. **(8 marks)**
(Total =20 marks)

SECTION B
Answer any 2 questions

4. **(a)** Define the following terms:
(i) Excretion. **(3, 2 marks)**
(ii) Osmoregulation.

(b) Draw a large labelled diagram of the mammalian skin. **(8 marks)**

(c) How does the skin function in temperature regulation during over-heating? **(7 marks)**
(Total =20 marks)

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5. **(a)** What is respiration?
(ii) List five (5) properties of the lungs that make it an efficient respiratory surface. **(4, 5 marks)**

(b) How does a molecule of oxygen in the atmosphere reach the lungs of a mammal? **(4 marks)**

(c) How would you demonstrate that germinating seeds evolve heat during respiration? **(7 marks)**
(Total =20 marks)

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6. **(a)** Define the following terms giving examples of each:
(i) Cell division.
(ii) Gene. **(3,3,3 marks)**
(iii) Homozygous.

(b) How is an insect pollinated flower different from a wind pollinated flower? **(6 marks)**

(c) How do mammals carry out parental care after birth? **(5 marks)**
(Total =20 marks)

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7. **(a)** Draw a well labelled diagram of the internal structure of a villus. **(9 marks)**

(b) State **six (6)** ways in which herbivorous mammals are adapted to their diet. **(6 marks)**

(c) What is the fate of the end products of photosynthesis? **(5 marks)**
Total =20 marks)

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